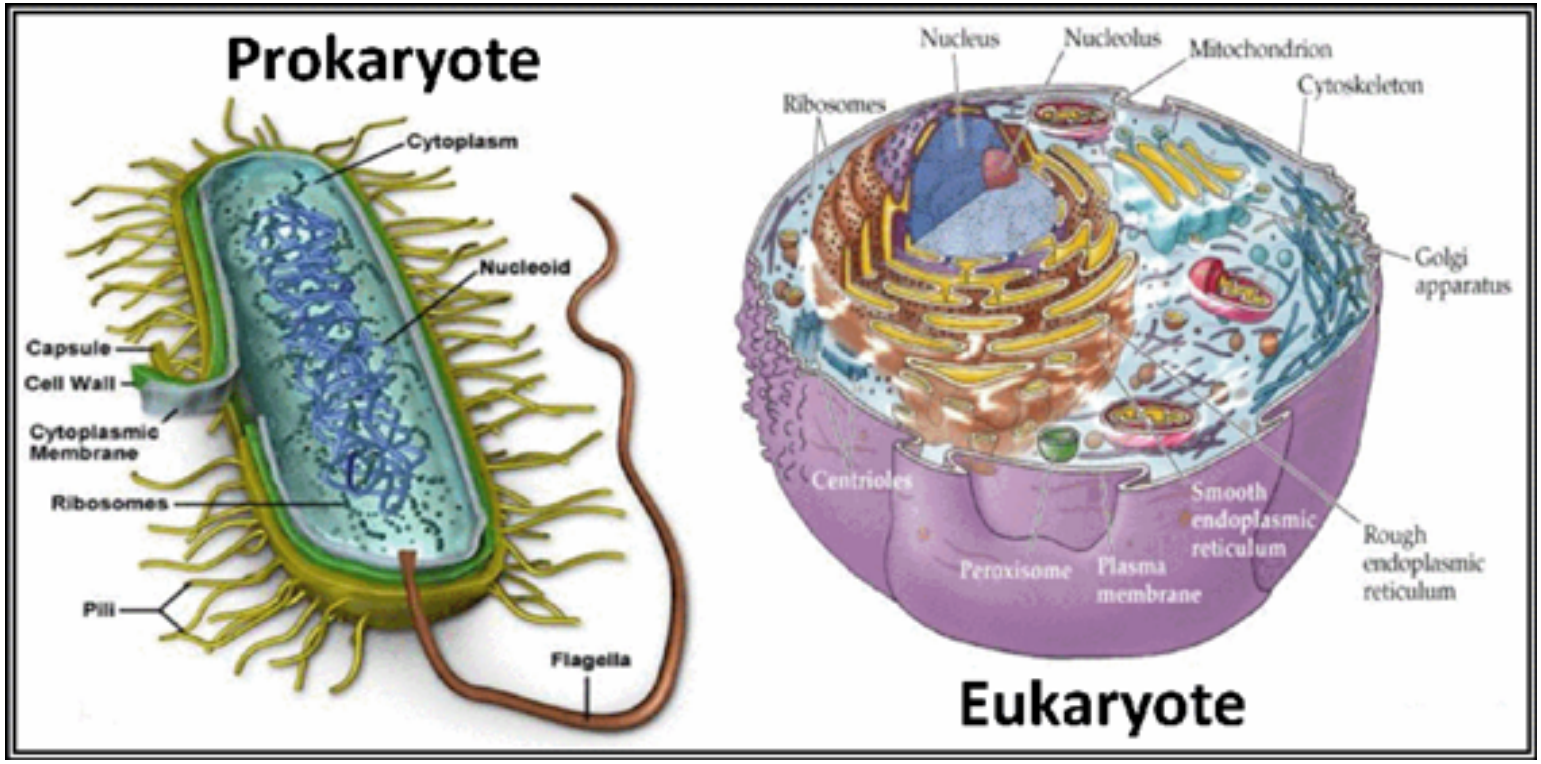
Computational Biology

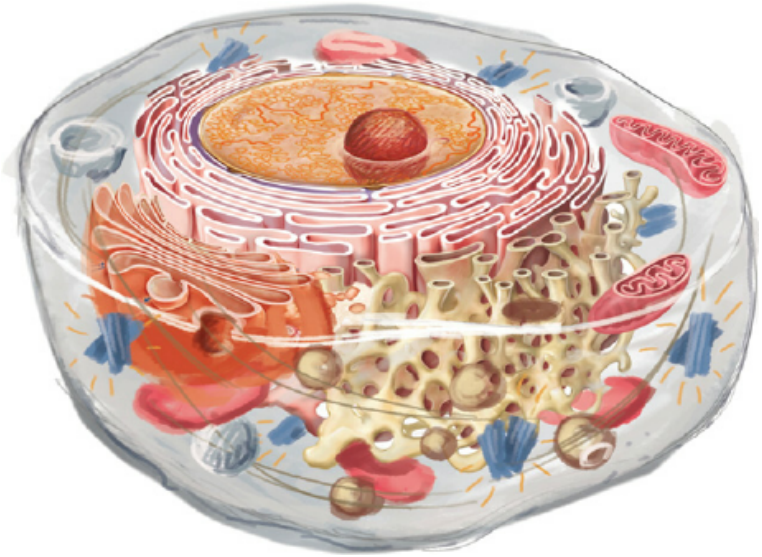
Lecture 2: Cell Regulation



Gene Regulation Networks

Outline

1. **Cells**
2. Protein Function
3. Cell Regulation



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- Cells are complicated machines
- Understanding how they work is still an on-going story
- We are going to concentrated on parts where computational biology has and is making an impact

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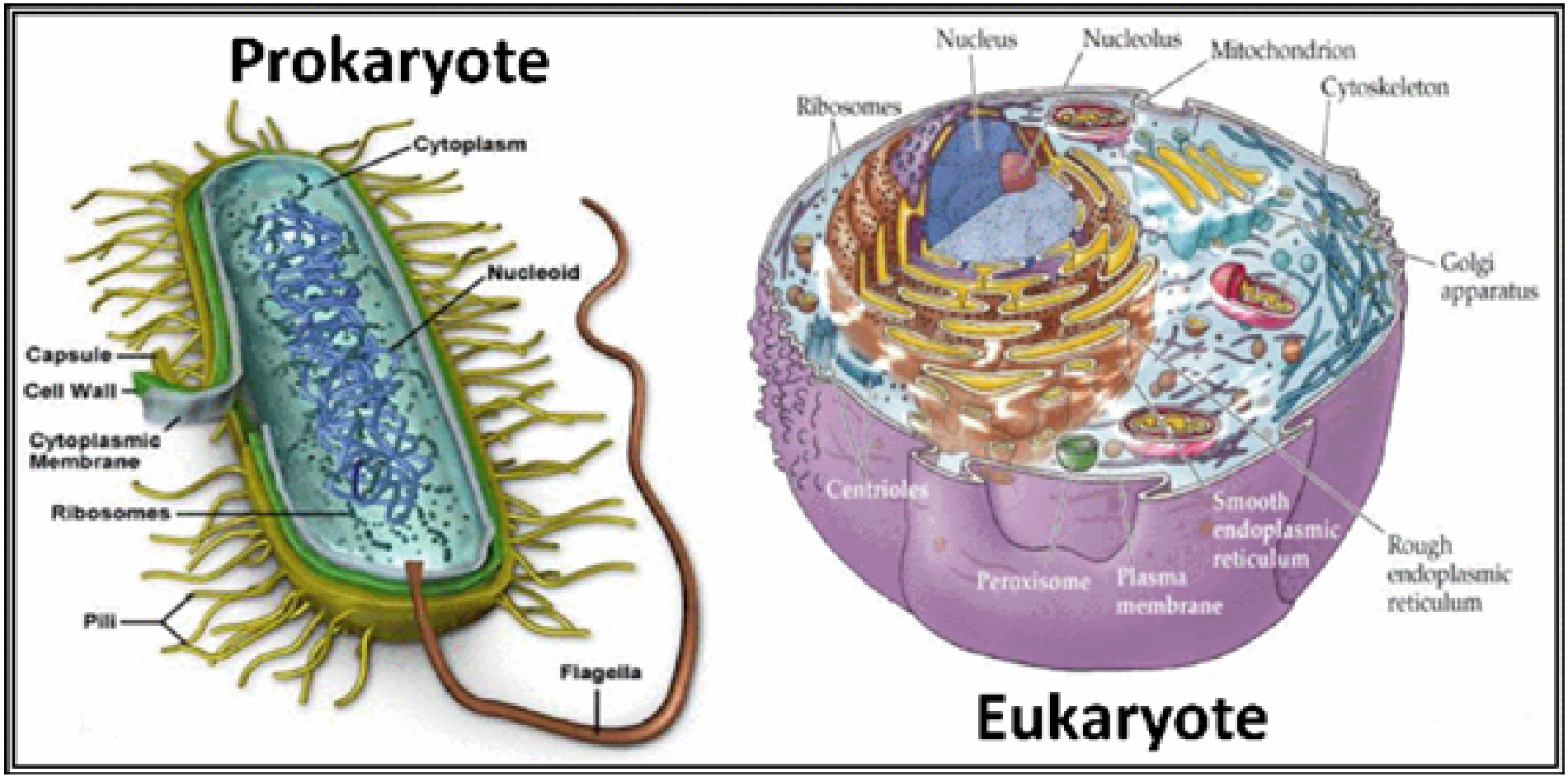

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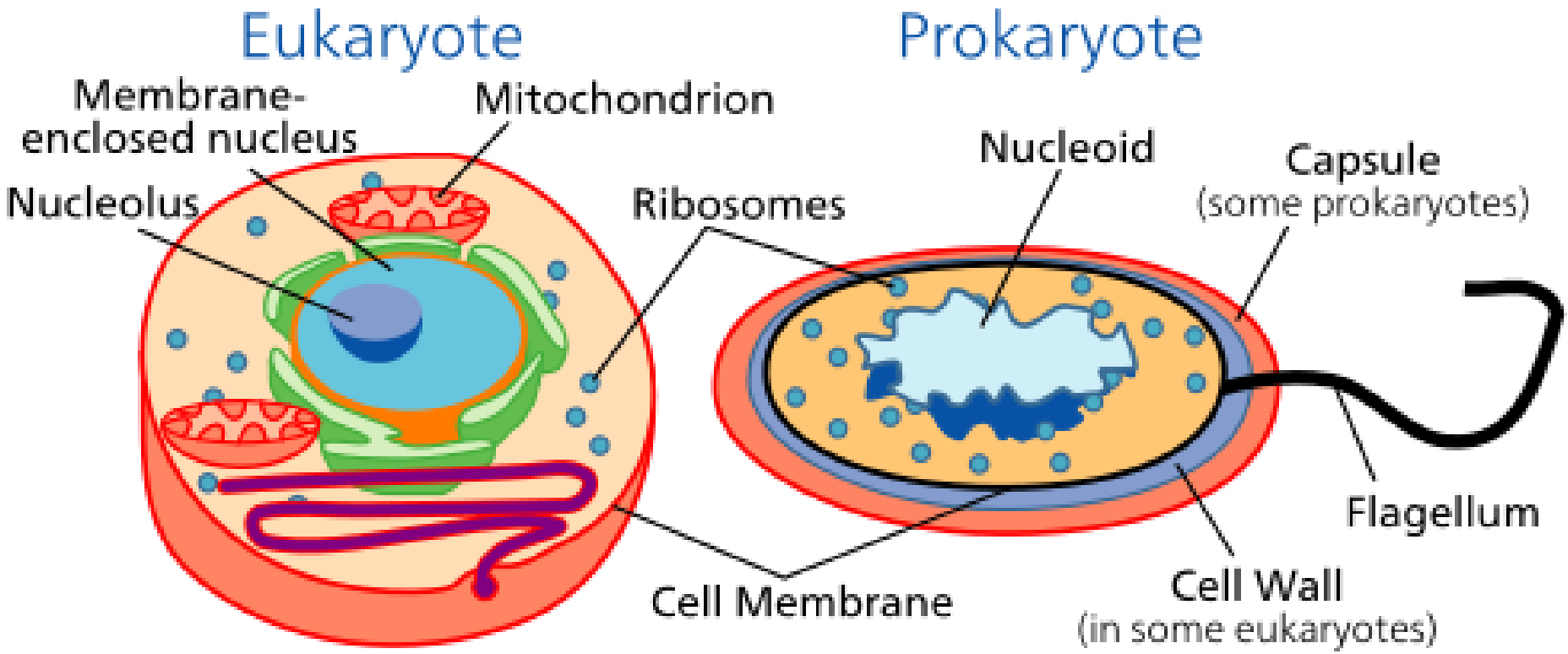

Differences and Similarities

- Share cell membrane, DNA code, ribosome (machine for producing proteins)
- Eukaryotes are much more complex
- It is now generally accepted that Eukaryotes were formed by fusing different cells
  - ⋆ nucleus
  - ⋆ mitrochondria (powerhouse, inherited from mother only)
- Eukaryotes can be multicellular

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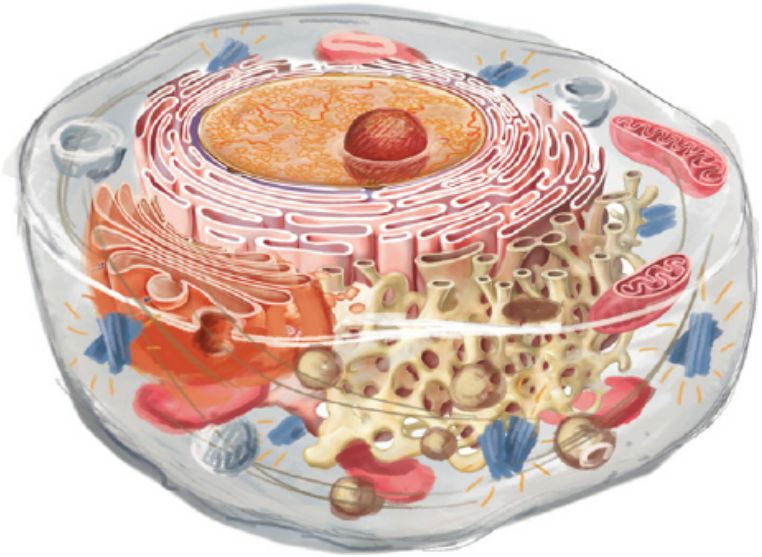

Understanding Proteins

- Proteins are hard to understand
- They are hard to sequence directly
- Fortunately we can deduce their sequence from DNA or mRNA
- Measuring protein concentration is difficult
  - ⋆ field of **proteomics** looks at this
  - ⋆ uses tools like mass spectrometer to count proteins
  - ⋆ beginning to do this much more often

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- To understand what proteins do we want to understand their three dimensional shape
- Alas this is hugely difficult
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- Predicting the structure is a non-local problem

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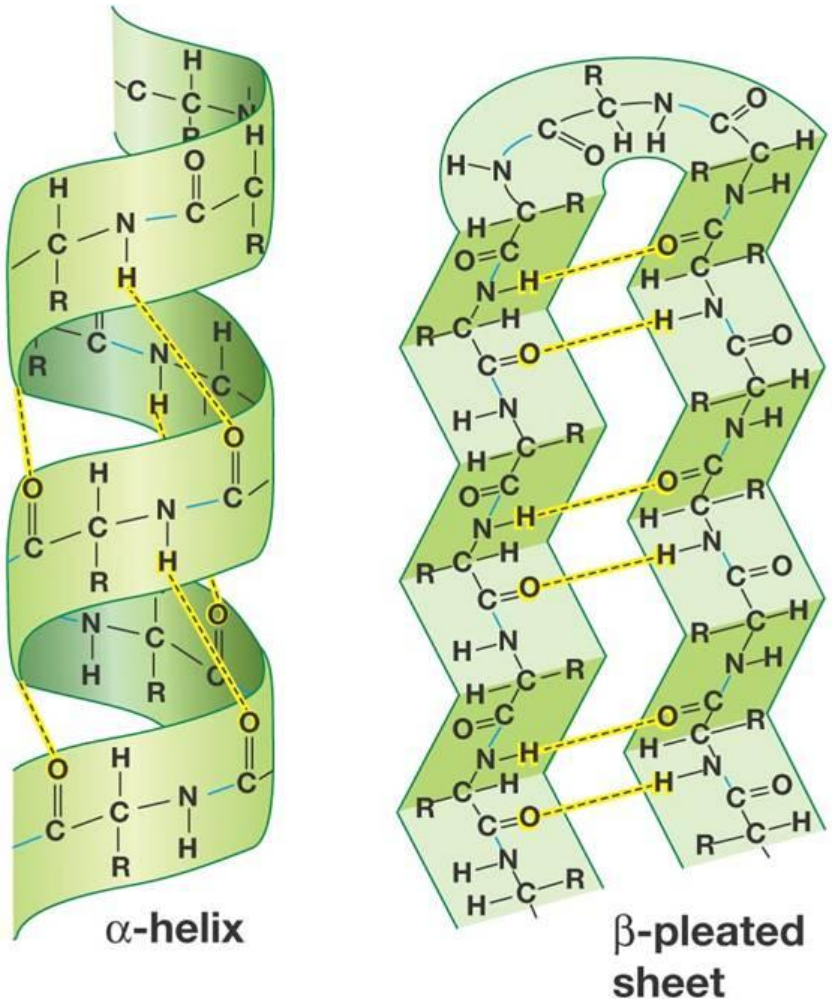
- Rather dramatically, this problem has almost been solved by alpha-fold

Secondary Structure



- The predominant components making up proteins are
  - ⋆ Alpha helices
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  - ⋆ Various folds and connecting parts
- This is known as the secondary structure
- We will look at secondary structure prediction later in the course

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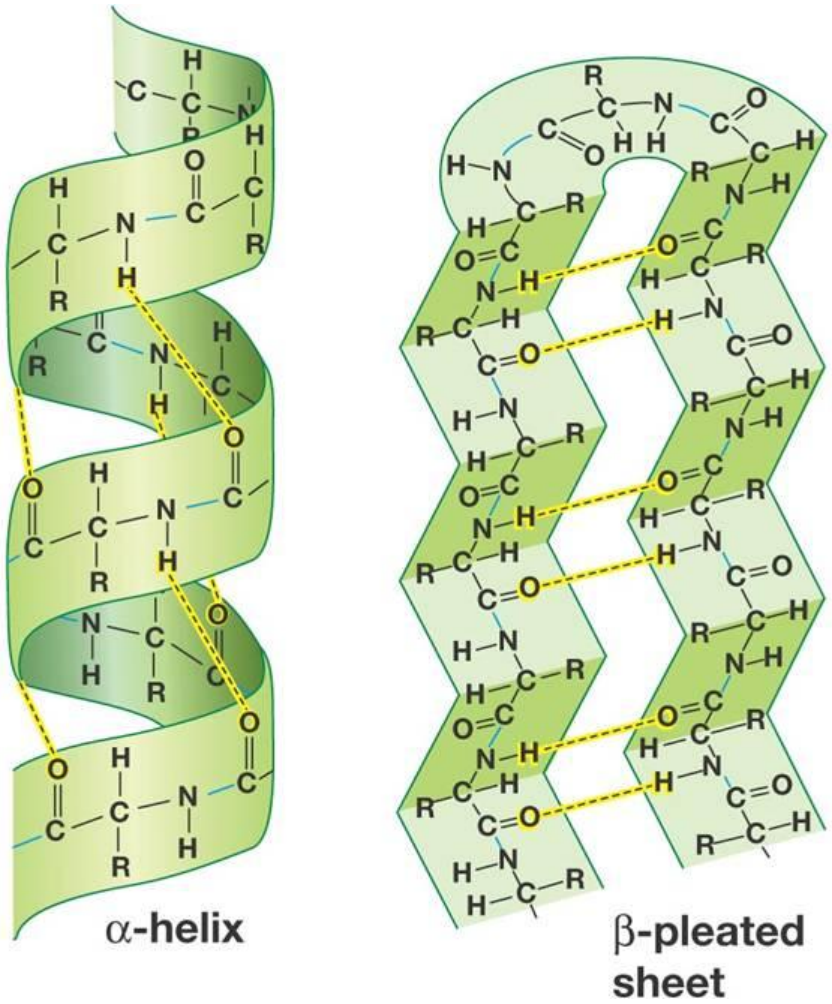

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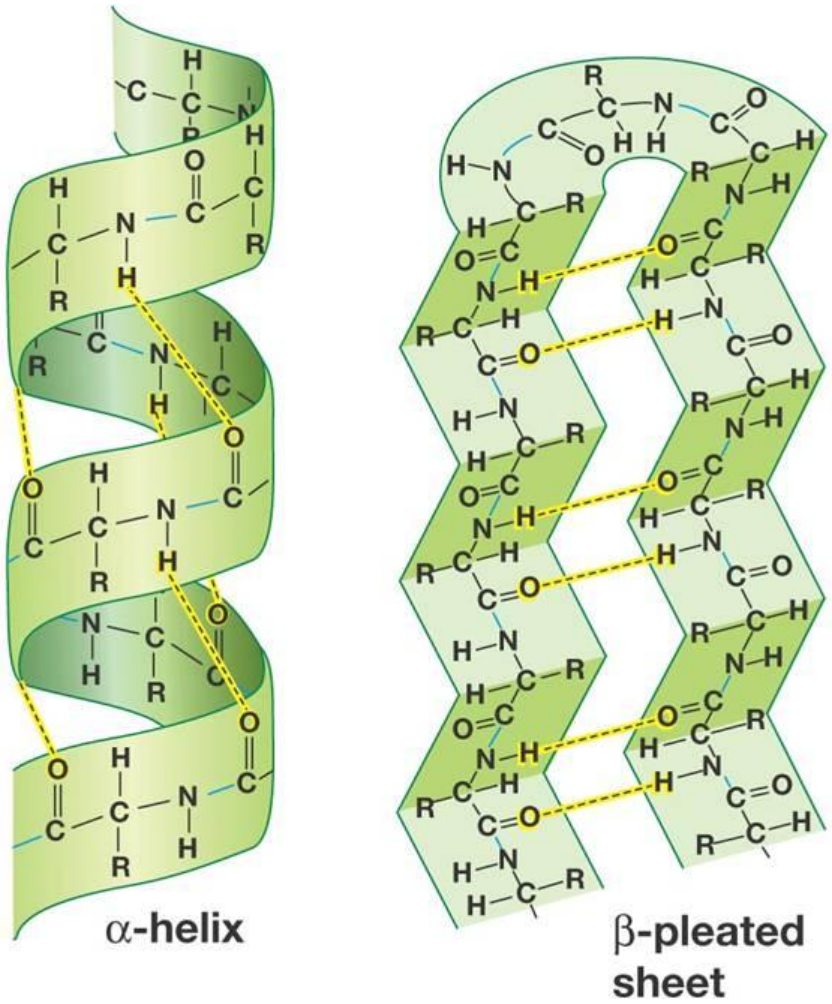

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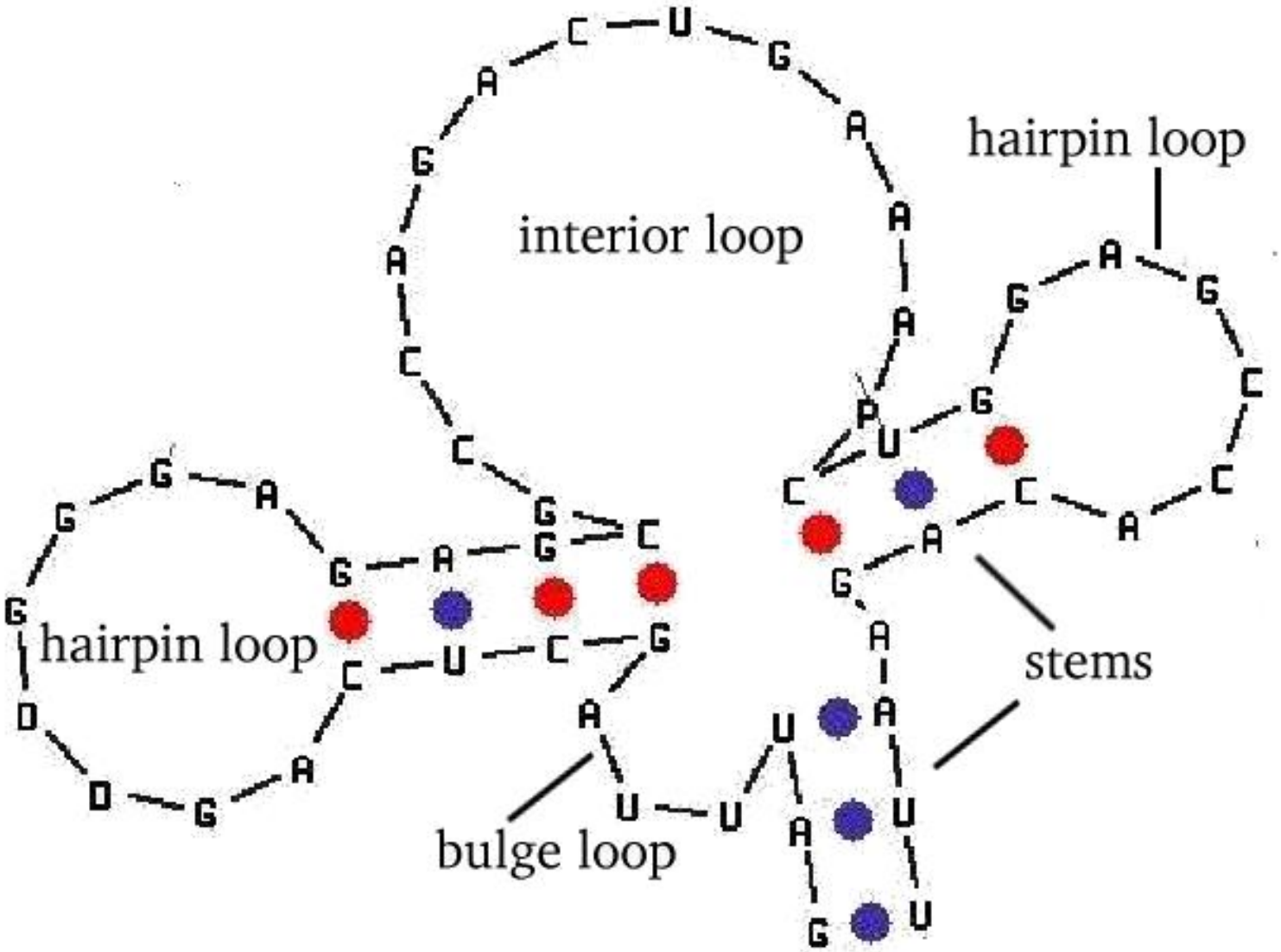

The First Law of Protein Sequences

Similar structures perform similar functions

- We can learn about the function of a protein by comparing with known proteins
- Because of evolution, many proteins are common to a huge number of organism
- Caveat: the converse is not necessarily true; dissimilar structures can have a similar function (convergent evolution)

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Sequence Alignment

Mouse *Pax6* gene:

GTATCCAACGGTTGTGTGAGTAAAATTCTGGGCAGGTATTACGAGACTGGCTCCATCAGA

Fly *eyeless* gene:

Genetic similarity to mouse: 76.66%
Protein similarity to mouse: 100%

GTATCAAATGGATGTGTGAGCAAAATTCTCGGGAGGTATTATGAAACAGGAAGCATACGA

Shark eye control gene:

Genetic similarity to mouse: 85%
Protein similarity to mouse: 100%

GTGTCCAACGGTTGTGTCAGTAAAATCCTGGGCAGATACTATGAAACAGGATCCATCAGA

Squid eye control gene:

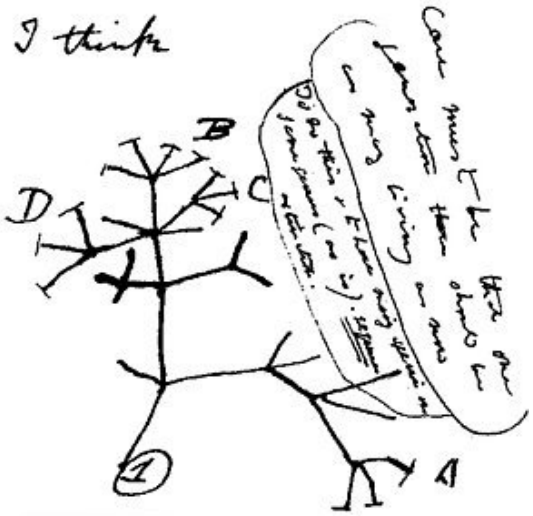
Genetic similarity to mouse: 78.33%
Protein similarity to mouse: 100%

GTCTCCAACGGCTGCGTTAGCAAGATTCTCGGACGGTACTATGAGACGGGCTCCATAAGA

Flatworm eye control gene:

Genetic similarity to mouse: 71.66%
Protein similarity to mouse: 100%

GTGTCTAATGGTTGTGTTAGTAAAATACTTTGCCGATATTATGGAACAGGTTCTATTAAA



Finding Aligned Sequences

- Biologists will routinely compare any new sequence with sequences already found
- This requires comparing with very large databases
- Typically sequences contain annotations indicating what someone believes a protein does in some organism
- We will spend sometime looking at alignment algorithms

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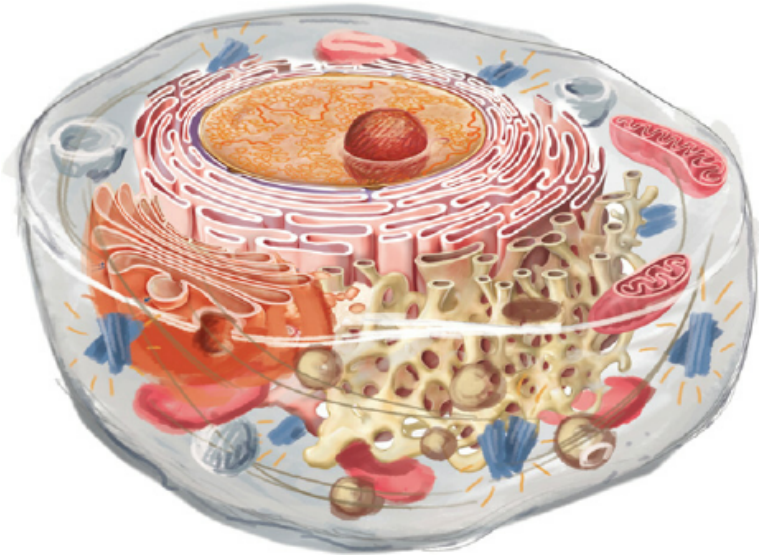

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- They therefore need a control mechanism
- We are at the beginning of understand these mechanisms
- Proteins and DNA lie at the heart of this

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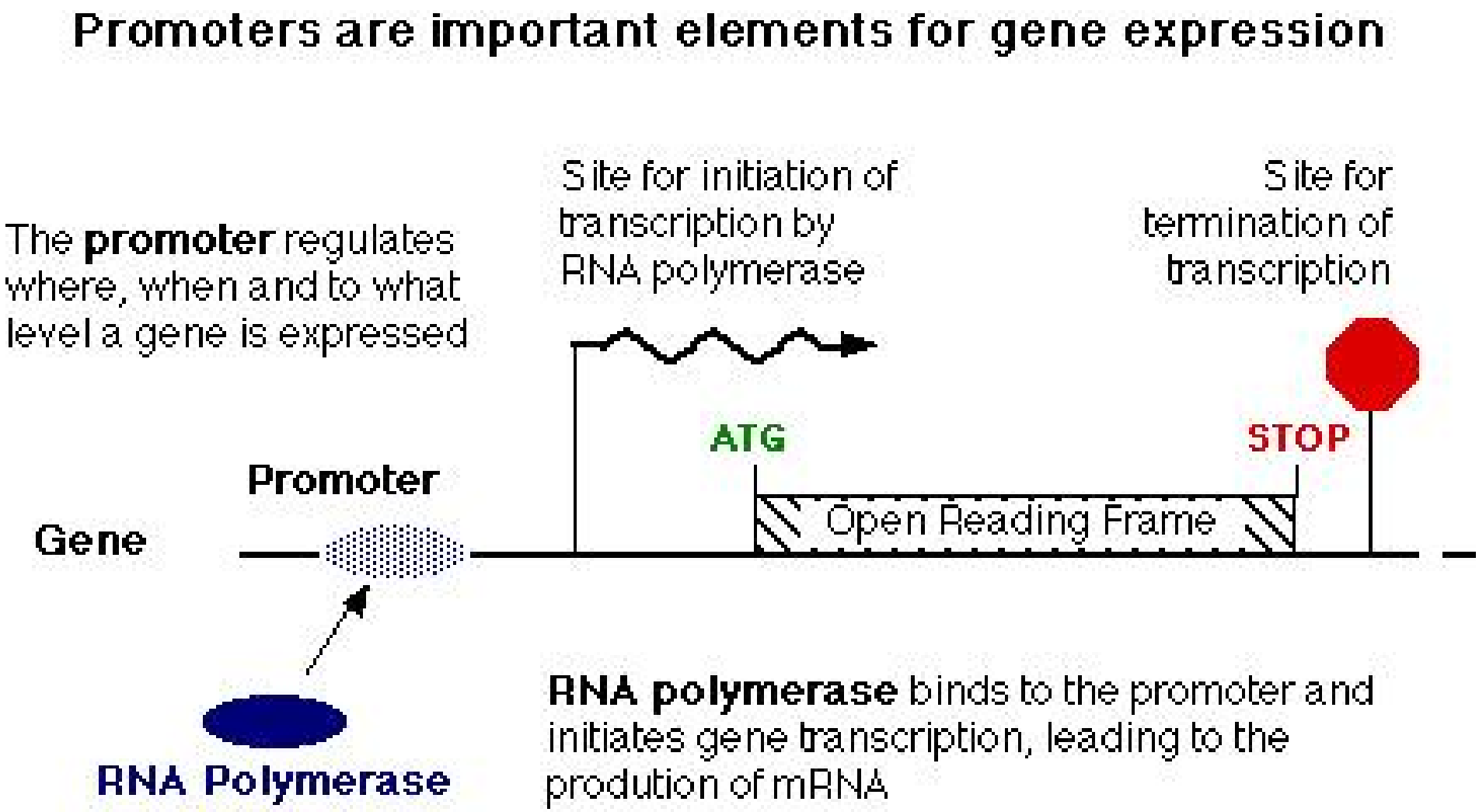
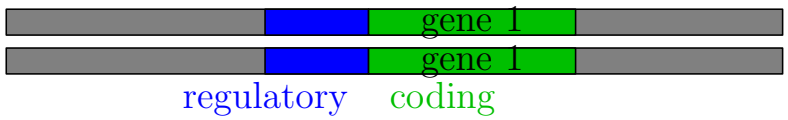
Promoter Region



Regulation by Signal



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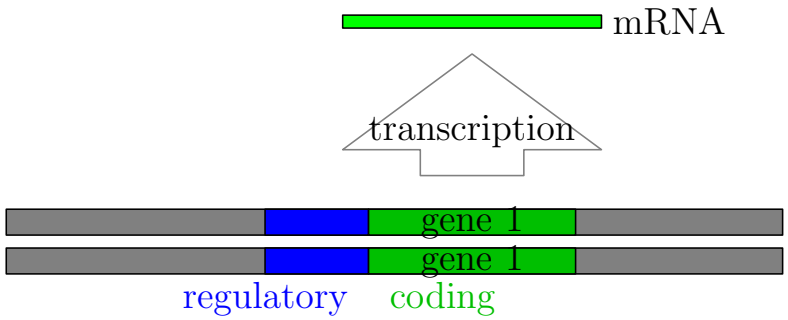

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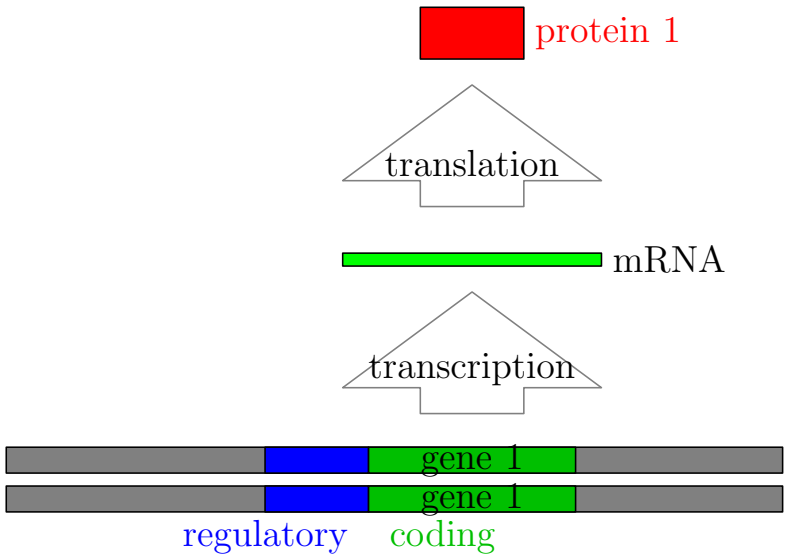

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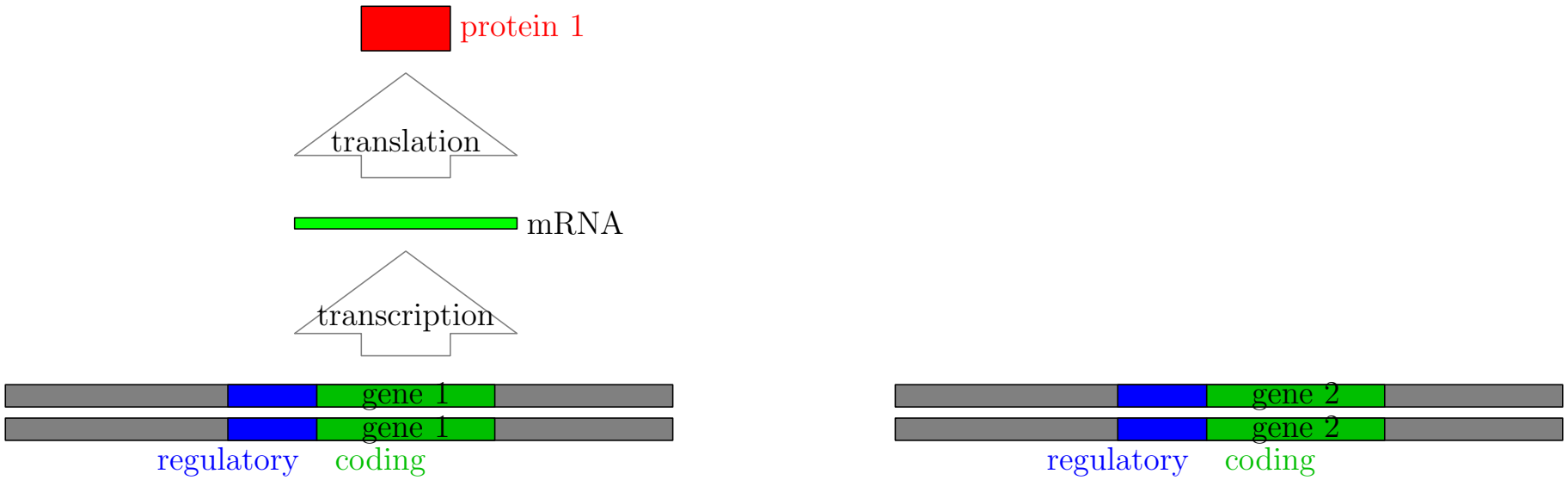

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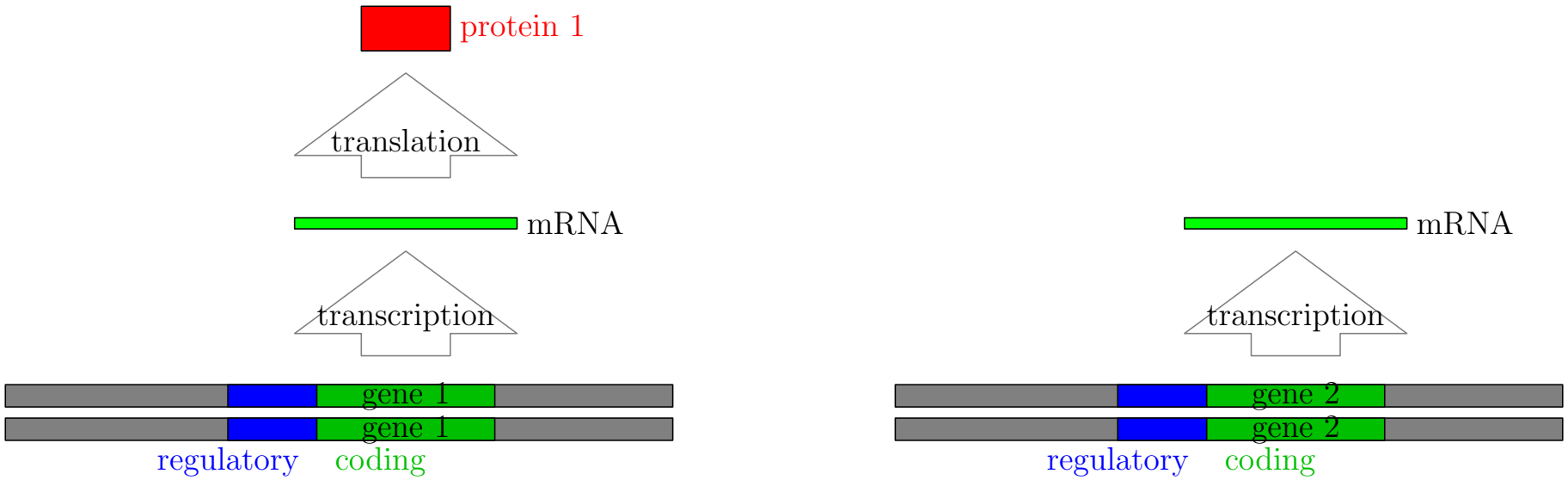

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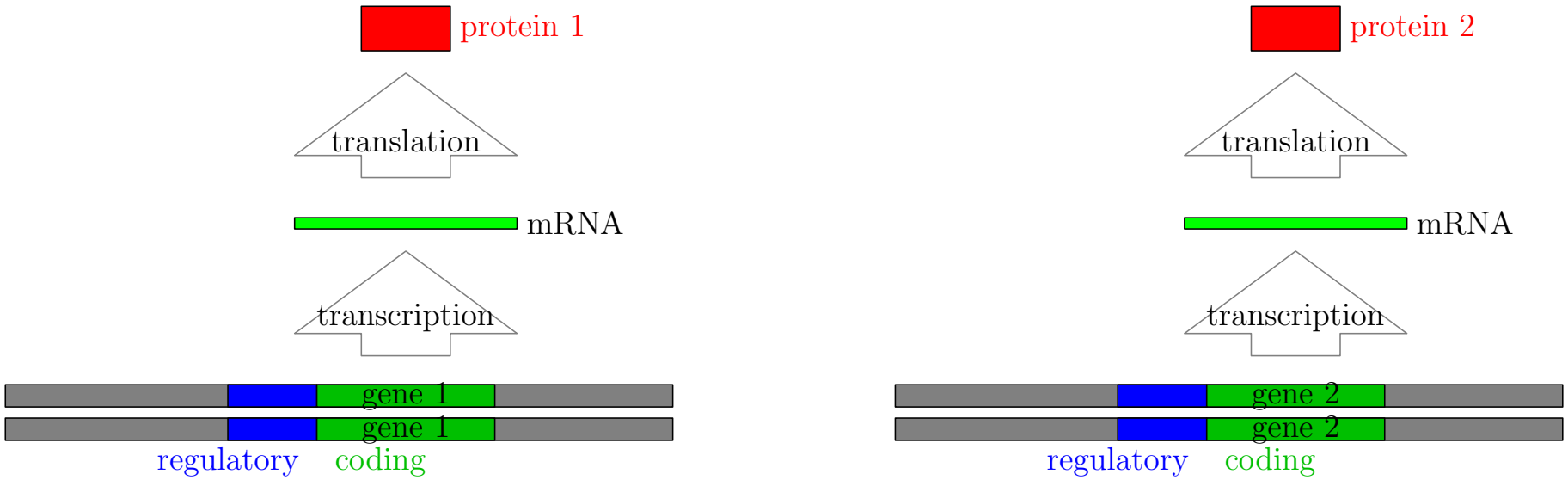

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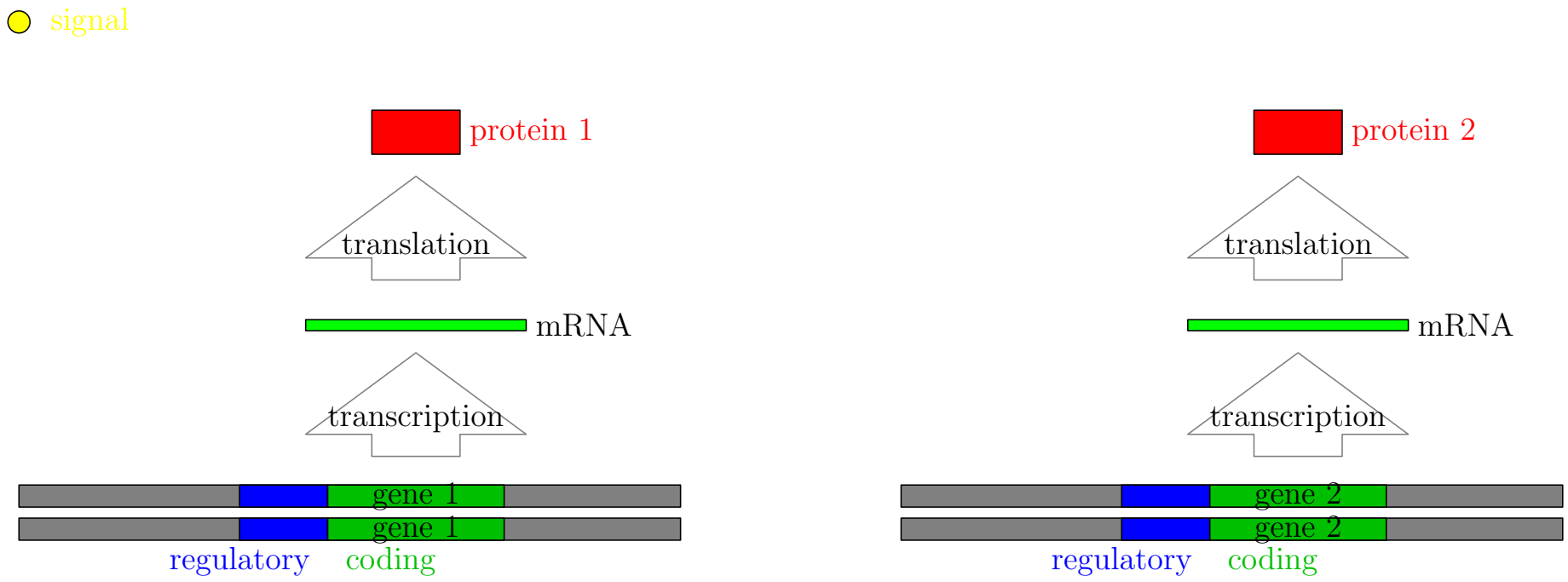

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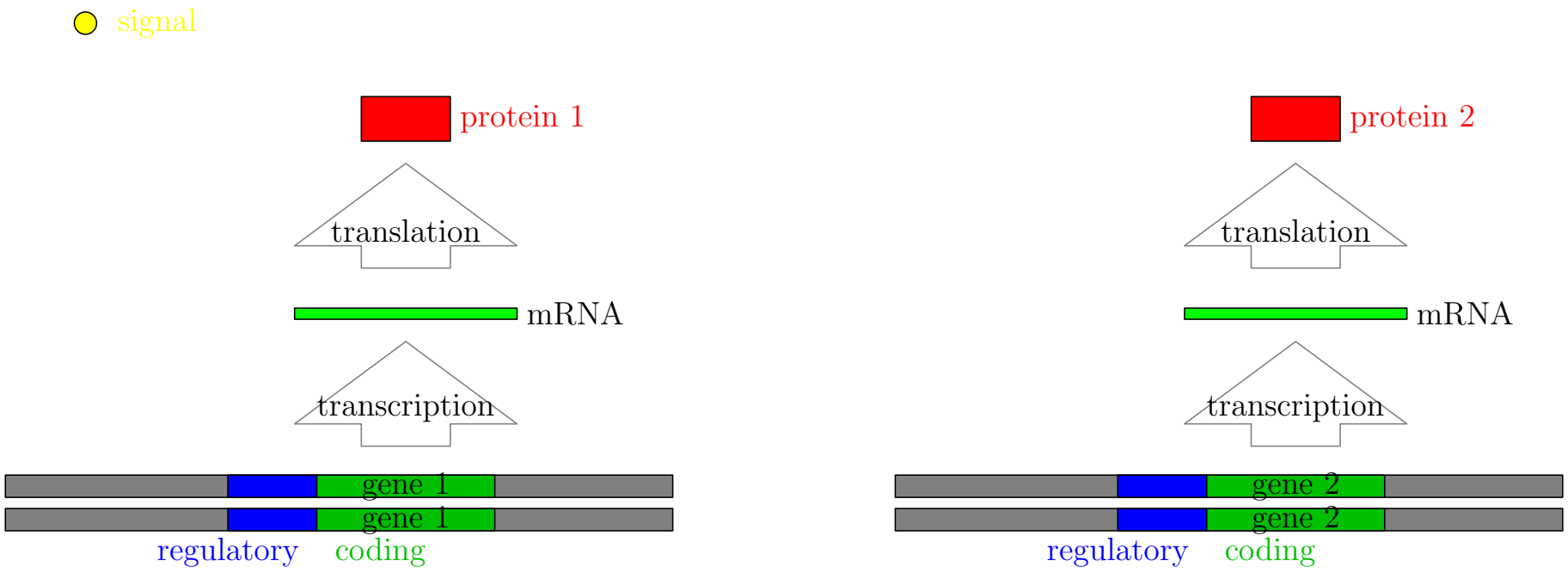

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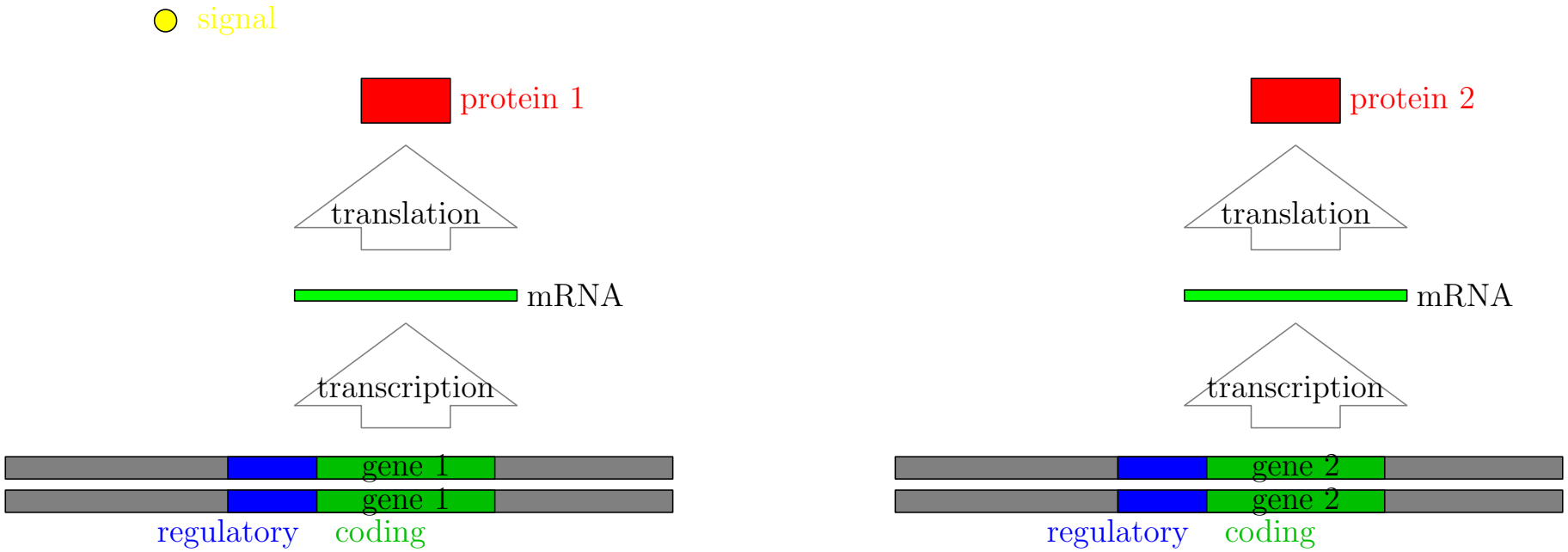

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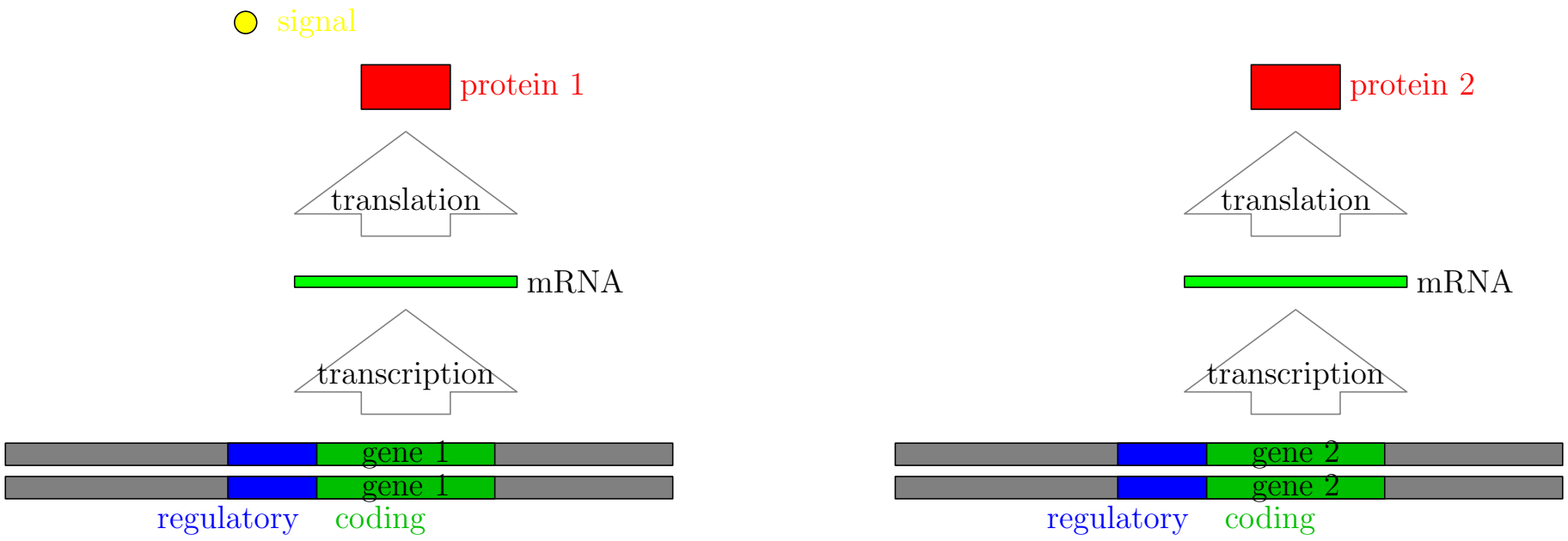

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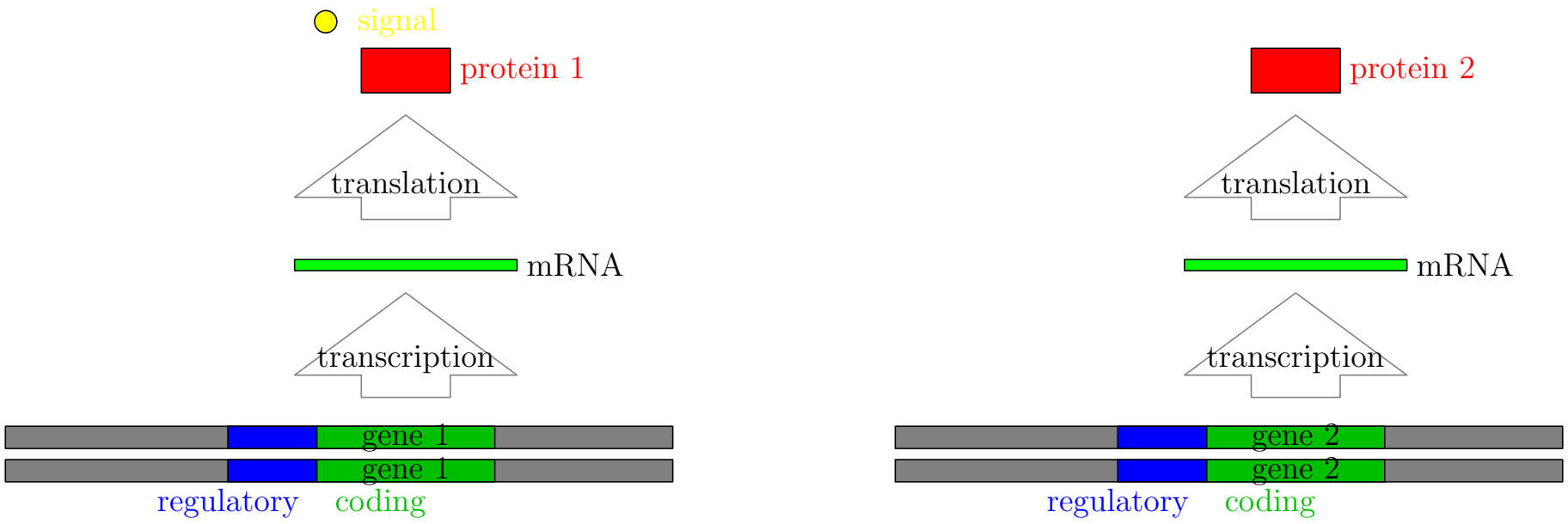

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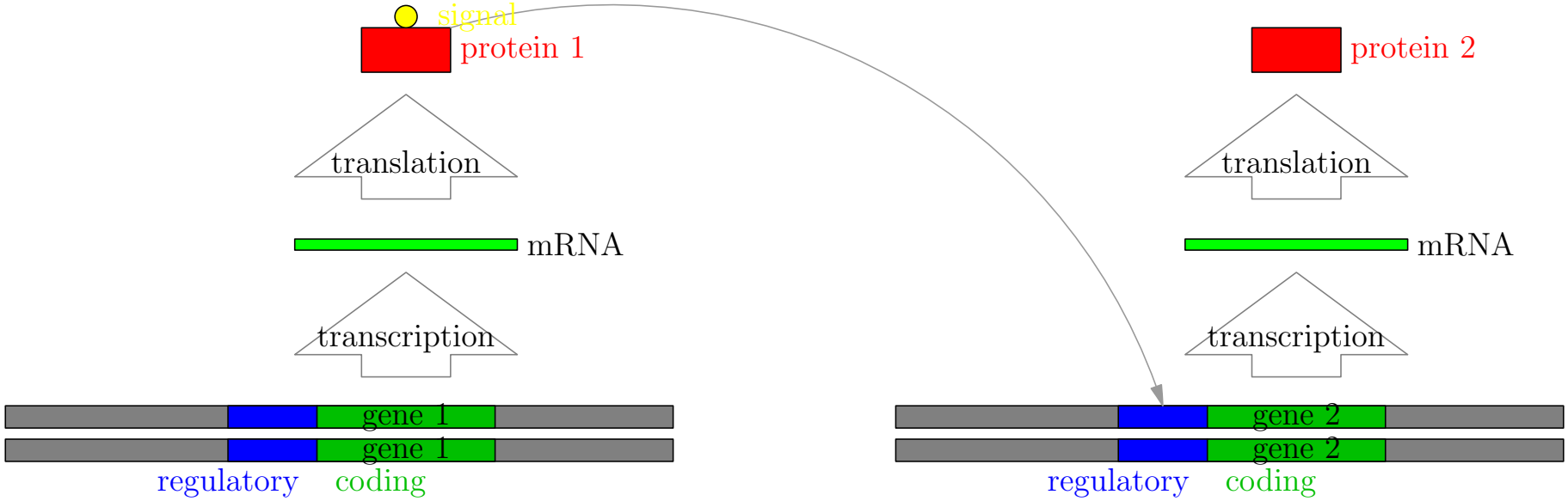

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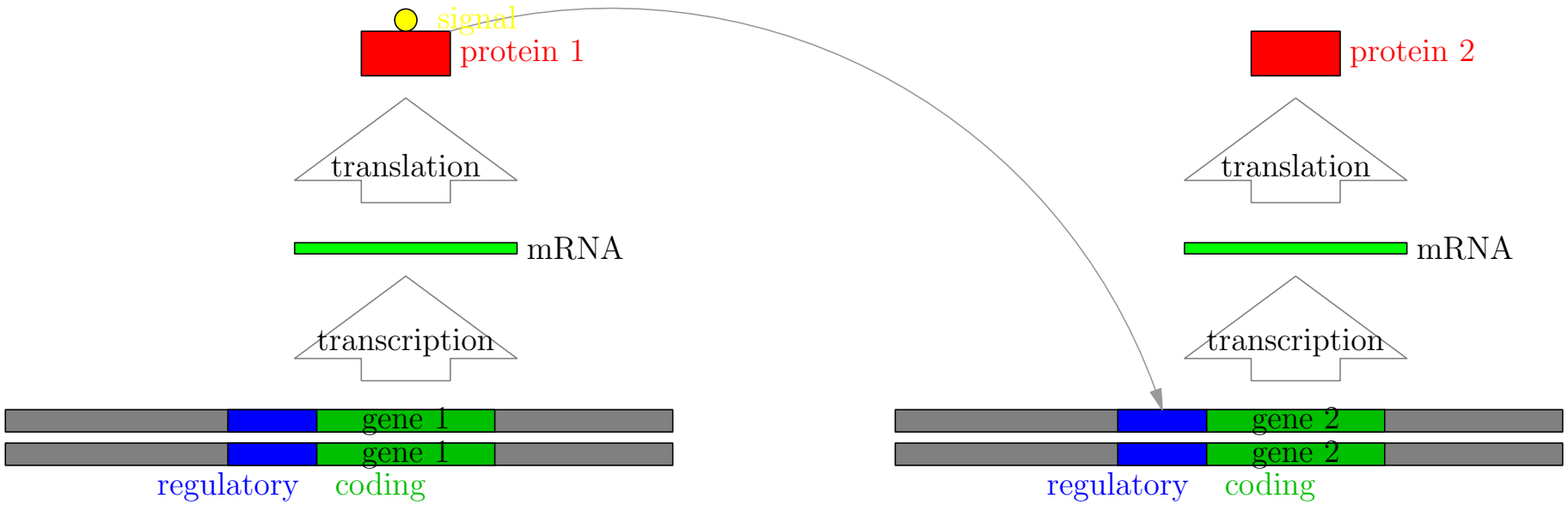
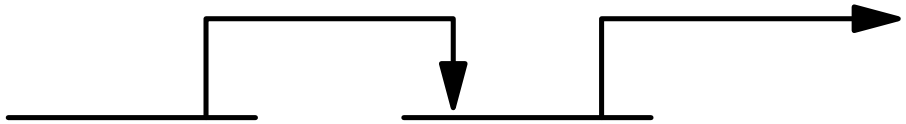


Excitation



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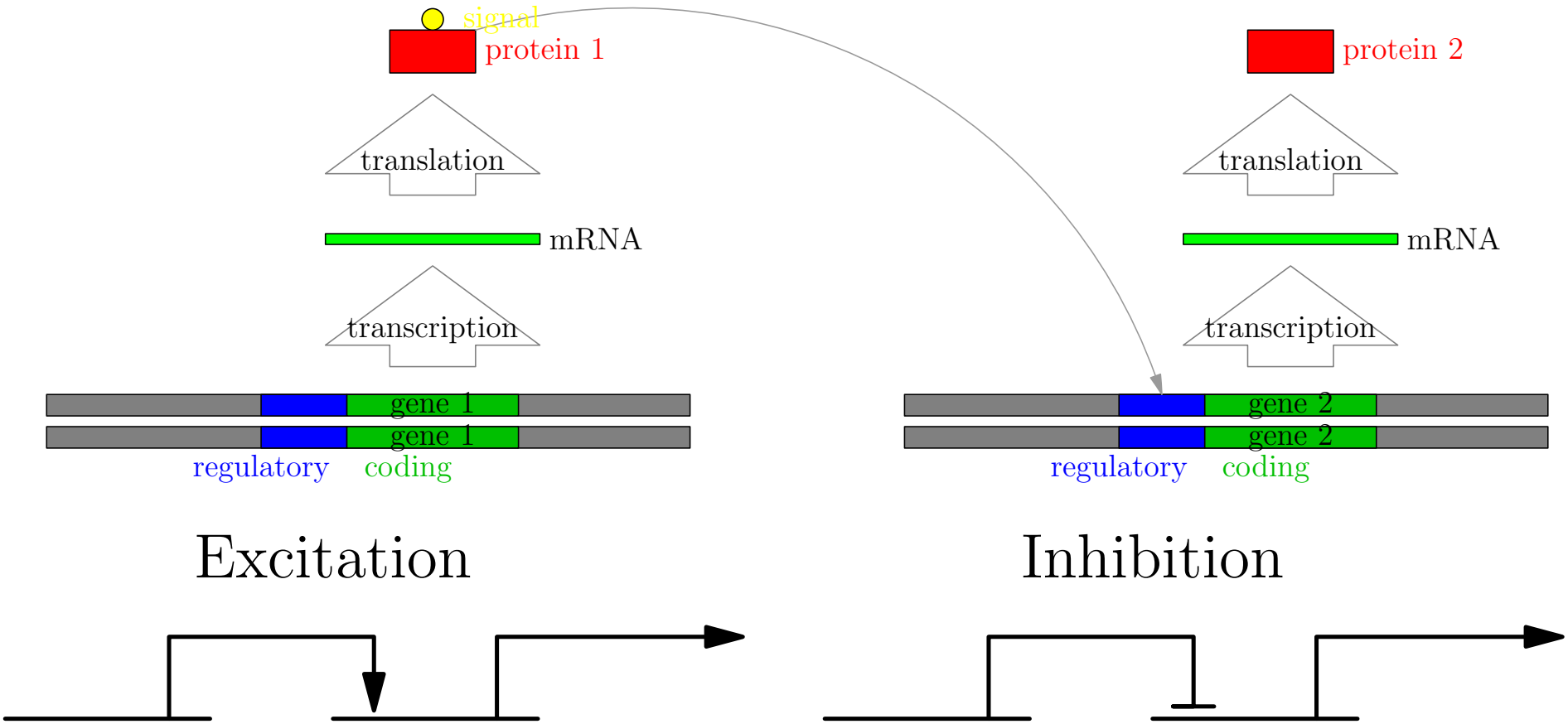

Regulation in Promoter Region

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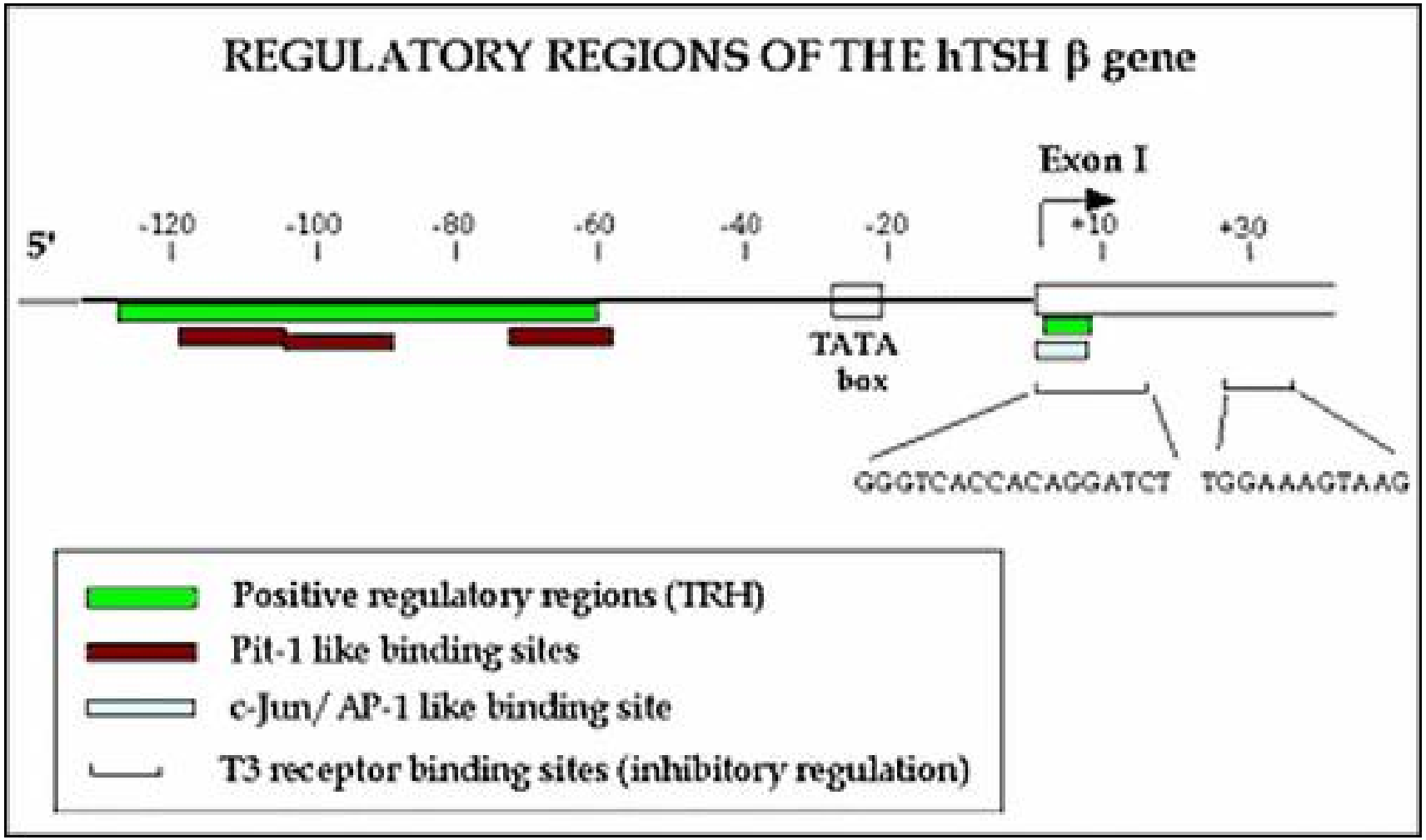
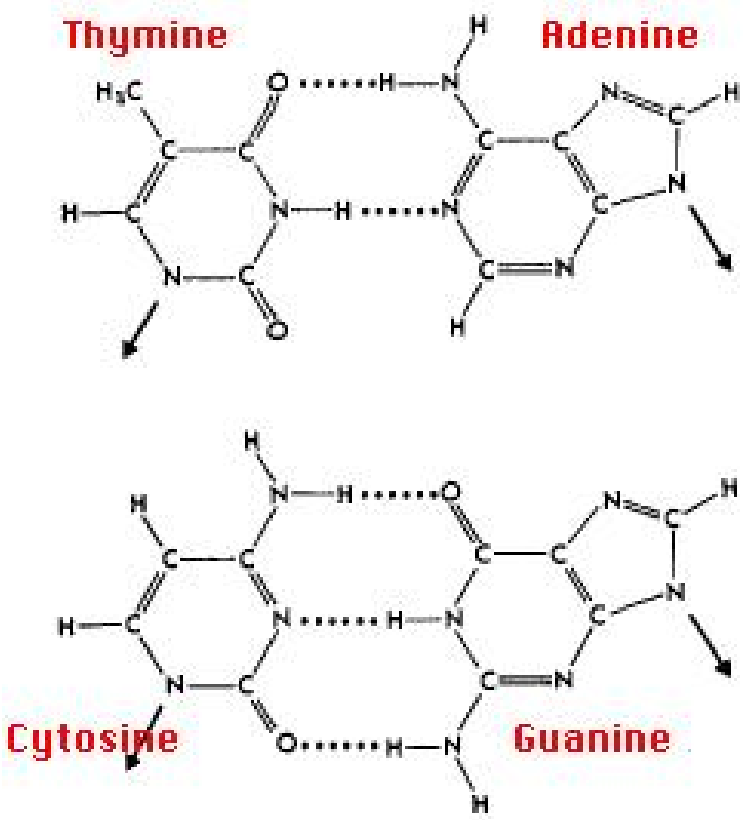


TATA box



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- Because A-T pairs have only 2 hydrogen bonds they are easier to break
- This helps RNA-polymerase split open the double helix

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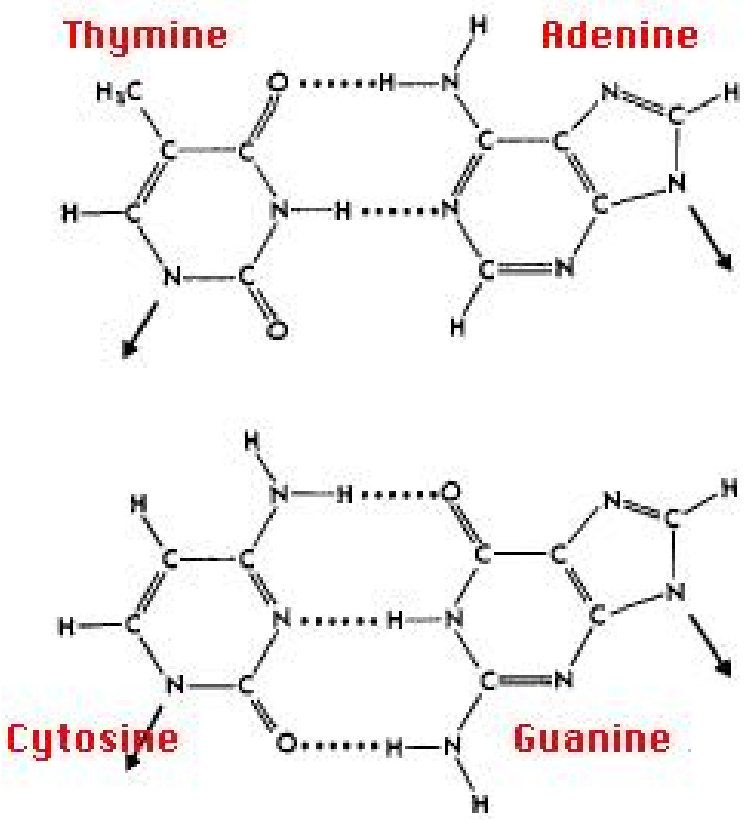

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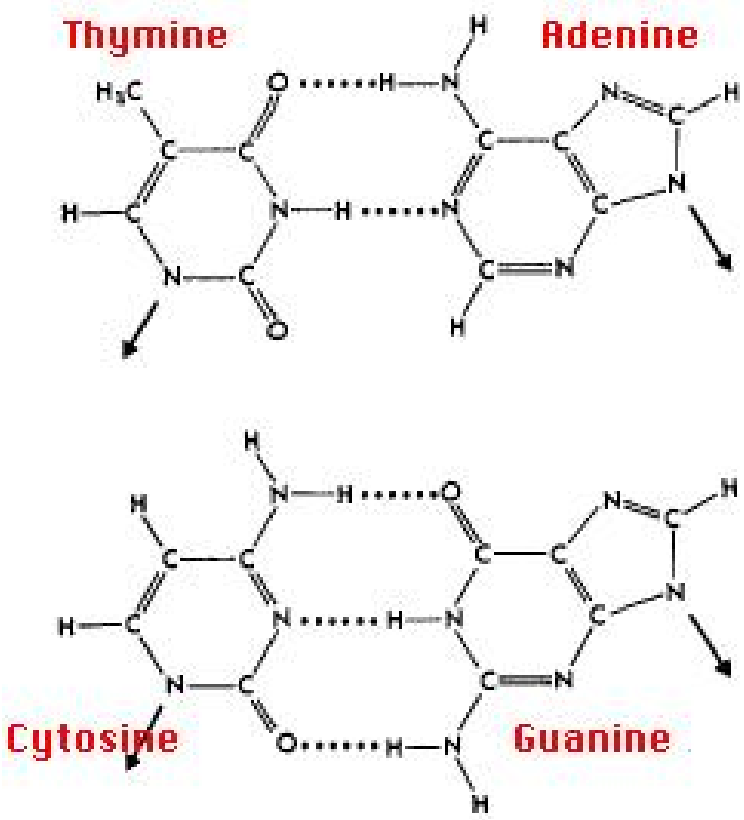

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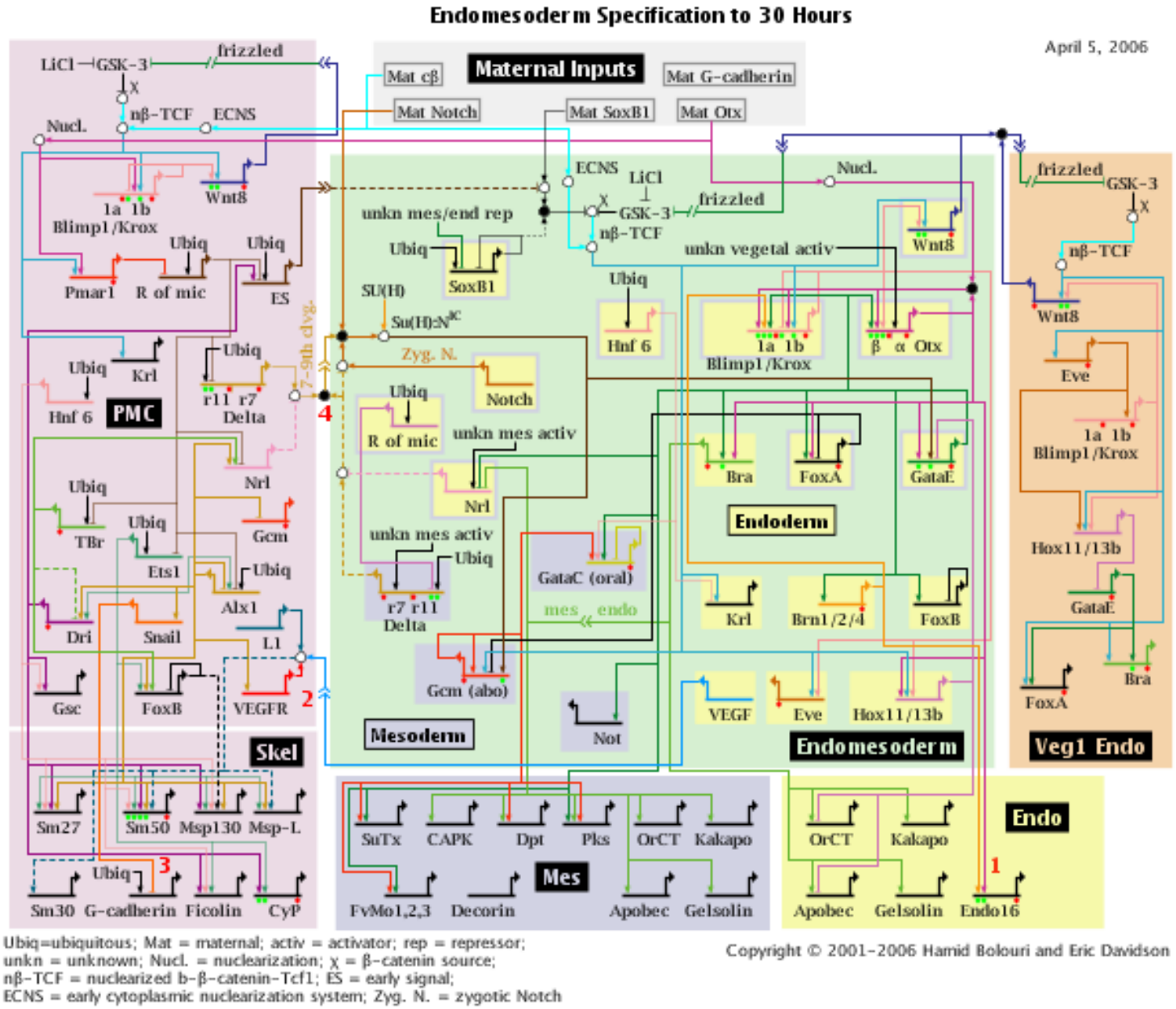
Sea Urchin Embryo GRN



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